

Utilization of environmental and epidemiological indicators in the study of malaria dynamics

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetórias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

The Amazon is one of the largest and most biodiverse tropical forests in the world, harboring numerous species of plants, animals, and microorganisms, including vectors and pathogens responsible for the transmission of various diseases. Among them, one of the most common is malaria, caused by protozoa of the genus *Plasmodium*, transmitted by the bite of the infected female mosquito of the genus *Anopheles*. It is present in 22 American countries, but the areas with the highest risk of infection are located in the Amazon region, encompassing nine countries, which accounted for 68% of infection cases in 2011 [1]. Although malaria is prevalent in the Americas, it is not limited to this continent and is found in countries in Africa and Asia, resulting in more than two million cases of infection and 445,000 deaths worldwide in 2016 [2], being endemic in a total of 84 countries in 2021 [3].

Notably, vector-borne disease transmission is closely related to environmental changes that interfere with the ecosystem of both transmitting and affected organisms. In the case of the Amazon, agricultural and livestock settlements are among the factors that most favor disease transmission, both due to the deforestation they cause for establishment and the clustering of people in environments close to the vector's habitat [4], especially by gathering non-immune migrants near these natural and artificial breeding sites [5].

Additionally, other factors such as rainfall, wildfires, and mining also significantly influence disease transmission in the region. These events result in habitat loss, ecosystem fragmentation, and climate changes, affecting the distribution and abundance of vectors and hosts, as well as their interaction with pathogens. Furthermore, population growth

and urbanization also play a crucial role in disease spread, increasing human exposure to vectors and infection risks.

In this context, this work aims to investigate vector-borne disease transmission in the Amazon and analyze how environmental impacts influence the dynamics of malaria transmission, the ecological and socioeconomic factors affecting this spread. This research was adapted from the Trajetórias Project, developed by the Center for Biodiversity and Ecosystem Services (SinBiose/CNPq), which is a dataset including environmental, epidemiological, economic, and socioeconomic indicators for all municipalities in the Legal Amazon, analyzing the spatial and temporal relationship between economic trajectories linked to the dynamics of agrarian systems, whether they are family-based rural or of large-scale agricultural and livestock production, the availability of natural resources, and the risk of diseases [6].

2. Model formulation

2.1. Base SIR-SEI model

Based on previous work developed to model the transmission of malaria based on precipitation and temperature dynamics [7], we initially focused on two different sets of compartments of hosts and vectors: susceptible, infected and recovered hosts (S_H , I_H , R_H respectively), and susceptible, exposed and infected vectors (S_M , E_M , I_M respectively).

The equations that describe the transmission are initially given as:

$$\frac{dS_H}{dt} = -ab_2 \left(\frac{I_M}{N} \right) S_H \quad (1)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) - \gamma I_H \quad (2)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (3)$$

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$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (4)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell(\tau_M) \end{aligned} \quad (5)$$

$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell(\tau_M) \\ & - \mu I_M \end{aligned} \quad (6)$$

The parameters are given in **Table A.1**, while the variables are given in **Table A.2**, which can be found in the Appendix.

However, the use of these equations proved to be unsuccessful in the appropriate modelling of the behavior of the disease, as can be seen in Figures 1A and 2A: while it would be expected for the number of infected humans to oscilate during the period of study, given that malaria is a recurring illness in the region, the model indicates that the number of infections would rapidly fall to 0, without occurrence of new cases.

In addition to that, it could also be seen that the mosquito population would also decline to near negligible levels, which could explain the lack of infections in the human population.

To overcome this effect, the SEI equations were modified in order for b_1 to only be used in the transference between compartments S and E , and a new parameter, b_3 to move mosquitos from E to I . Being inversely related to the incubation period, the definition of this rate can be found in **Table A.2**. In addition to that, the parameter $a(T)$ that was in use in the transition between exposed and infected also had to be removed, as no further bites occur at this point. The parameter l also had to be disassociated from the infection rate of exposed individual (b_3).

The SIR equations were also modified, to take into consideration population dynamics, including a r_H parameter as the net human increase rate per capita, which could be explained by births or immigration to the municipality. Given the positive growth of the studied population, a death rate was neglected for this study.

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M}{N} \right) S_H \quad (7)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M}{N} \right) S_H - \gamma I_H \quad (8)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (9)$$

$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (10)$$

$$\frac{dE_M}{dt} = ab_1 \left(\frac{I_H}{N} \right) S_M - (\mu + b_3 + l) E_M \quad (11)$$

$$\frac{dI_M}{dt} = b_3 E_M - \mu I_M \quad (12)$$

2.2. Base SEIRS-SEI model

Preliminary analysis of the baseline SIR-SEI framework revealed structural limitations in replicating endemic malaria transmission, as the system intrinsically tended toward a single outbreak followed by rapid fade-out. To rectify this and enhance biological fidelity, we extended the model to a SEIRS-SEI structure. This extension accounts for essential features of Plasmodium dynamics, such as the extrinsic incubation period in humans, represented by the Exposed compartment (E_H), and the decay of temporary acquired immunity, represented by the transition from Recovered to Susceptible ($R_H \rightarrow S_H$). These modifications are necessary to sustain the multi-year recurrence and seasonal oscillations characteristic of endemic regions.

The constructed model is as follows:

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M(t)}{N} \right) S_H + \omega R_H \quad (13)$$

$$\begin{aligned} \frac{dE_H}{dt} = & ab_2 \left(\frac{I_M(t)}{N} \right) S_H \\ & - ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{dI_H}{dt} = & ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \\ & - \gamma I_H \end{aligned} \quad (15)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (16)$$

$$\frac{dS_M}{dt} = \Lambda - \left(ab_1 \left(\frac{I_H(t)}{N} \right) + \mu \right) S_M \quad (17)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H(t)}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \\ & - \mu I_M \end{aligned} \quad (19)$$

Where Λ is the mosquito recruitment rate:

$$\Lambda = b(R, T) \cdot K \cdot \left(1 - \frac{S_M + E_M + I_M}{K} \right) \quad (20)$$

The updated equations used in the model are as following:

And K is the mosquito carrying capacity:

$$K = M \times \text{Temp. factor} \times \text{Rain factor} \times \text{Seasonal factor} \quad (21)$$

Where:

$$\text{Temp. factor} = \max \left(0.1, \min \left(1, \frac{T - 15}{15} \right) \right) \quad (22)$$

$$\text{Rain factor} = \min \left(1, \max \left(0, \frac{R}{30} \right) \right) \quad (23)$$

$$\text{Seasonal factor} = 2.5 + 2 \sin \left(2\pi \frac{t - 180}{365.25} \right) \quad (24)$$

$$(1 \leq t \leq 366)$$

3. Data description and processing

It was decided to use data from cases caused by *P. vivax* as this species is responsible for the highest number of malaria cases in Brazil [11, 12], being the most common human malaria parasite in the region [13].

Based on higher incidence and number of cases, the municipality initially chosen for analysis was of Manaus, especially the rural zones, as it had the highest incidence of malaria caused by *P. vivax* in the Amazon region. Now, as it was not possible to identify the population in rural areas based on available census data, we used the Trajetórias Project to estimate this population based on the incidence function given [6]:

$$\text{Inc} = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \times 5 \text{ years}} \times 10^5, \quad (25)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ is the number of cases of disease d in zone z of municipality m , and t_1 and t_2 are the initial and final years of the interval, while $\text{Pop}(m, z, (t_1 + t_2)/2) \times 5 \text{ years}$ is the population in zone z of municipality m in the middle of the period multiplied by the total number of observation years. In the case of the rural population, we could indicate it as:

$$\text{Inc}(\text{Viv.}, \text{Man.}, \text{Ru.}, 2004, 2008) = \quad (26)$$

$$= \frac{\text{Cases}(\text{Viv.}, \text{Man.}, \text{Rur.}, 2004, 2008)}{\text{Pop}(\text{Man.}, \text{Rur.}, 2006) \times 5 \text{ years}} \times 10^5 \quad (27)$$

$$\Rightarrow 184030.8 = \frac{78745}{5\text{Pop}} \times 10^5 \Rightarrow \text{Pop} \approx 8558 \quad (28)$$

Using data on the total population of Manaus from 2004 to 2008, with an incidence of 3106.429047 and a number of cases of 262,264, the total population of the municipality was estimated to be 1,688,524 inhabitants, which corresponds to the value found in the official census [14]. Thus, the rural population represented approximately 0.5% of the municipality's population in this period, while the number of cases in rural setting is about 30% of all cases of malaria

caused by *P. vivax* in Manaus in this period, and about 9.4% of the cases in the state.

Using data from 2015 to 2019, the rural population is found to represent approximately 0.4% of the municipality's population, the number of cases in rural settings being about 41.5% of all cases in the city in this period, and about 7.2% of the cases in the state. With this, we will use that the rural population represents roughly 0.45% of Manaus' population, and the number of rural cases to be around 35.75% of the total cases in the city.

The yearly population was constructed via interpolation using data from IBGE (Brazilian Institute of Geography and Statistics, governmental body responsible for the official Brazilian censuses), based on the censuses of 2000, 2010 and 2022, and an estimation of the population of 2025 [15, 16, 17].

As the data on malaria cases from the Trajetórias project was not given annually, but instead as a sum of cases in each five-year period of study, yearly data was obtained from SIVEP [18], which registers notifications from every brazilian municipality from 2003 to 2023.

In order to use the latest and complete available data on registered cases and deforestation and forest fires noticed, this study was done using data from 2016 to 2024, but limited to the years of 2017 to 2023, given that the SIVEP database didn't include registered data from 2024 at the time of development of this study.

4. Impacts of environmental factors

As for the theory behind environmental factors, it is known that the removal of tree canopies allowed the resurgence of malaria in South America [20]. In deforested areas, without tree canopies covering the ground, water puddles under sunlight attract mosquitoes of the species *Anopheles darlingi*, the main vector related to human malaria in the Amazon [21]. They are usually less commonly found in still intact forests. This is because light and heat favor the development of larvae and pupae, in addition to a greater availability of algae for larval feeding [22]. The increase in ambient temperature also favors the vectorial capacity of mosquitoes.

Deforestation also attracts and brings humans closer to take part in logging, agriculture, and road construction activities, agglomerating individuals infected with *Plasmodium* to an area where both the vector and the environment have already been modified to favor transmission. Furthermore, agriculture also promotes river sedimentation, providing suitable

environments for breeding sites. Therefore, it can be considered a relevant change for the model to take into account deforestation, the increase in survival probabilities of eggs, larvae, and pupae, as well as increasing the proportion of bites that lead to infection, due to the increased human population density in areas near mosquito breeding sites.

It was outside the scope of this project to collect samples of *A. darlingi in loco*. Because of this it was instead used an estimate for the number of mosquitos based in literature. As the study was made around the rural population of the municipality, this would favour the clustering of adult mosquitos, given the proximity to bodies of water and recently open up areas of rainforest [23]. In fact, the municipality of Manaus has a great number of breeding sites in its periurban and rural areas, both natural and man-made. Many of the city's fish farming tanks have been noted as breeding sites for *A. darlingi* [24].

As a means for the development of infrastructure, agricultural expansion or other forms of occupation of the land, deforestation impacts the biodiversity in the region, changing the microclimate, water resources and the distribution of infectious diseases [25, 26], those transmitted by mosquitos being especially sensitive to changes in the habitat, suitability of pathogen development and contact between vector and host.

It is expected that the process of forest fragmentation would favour the development of larva by increasing light and temperature over settlements and habitats, leading to a shorter development time from larva to adult, a greater availability of algae for larval feeding, as well as increasing the vectorial capacity of mosquitos and their survivorship [27, 28].

As a method used in particular for land-clearing, forest fires remove the vegetation [29, 30], growing the risk of future fires by increasing the availability of combustible and understory drying [31]. This process became more common from 2019 to 2022, with simultaneous fires being set in the Amazon region, and in 2024 the state of Amazonas has reached it's highest number of recorded outbreaks since 1998, with over 21,600 cases reported, and over 10,000 just in the month of August [32, 33].

Monthly data of forest fires and yearly data of deforestation given in hectares of forest supression between 2017 and 2021 was obtained from MapBiomas [34, 35]. Monthly data for deforestation in Manaus was approximated from notifications from the state of Amazonas during this time interval, and it was estimated that 3% of notices of deforested area (km²) in the state were located in Manaus [36, 34]. In addition to that, daily notices of deforestation were also used, obtained from TerraBrasilis [37].

5. Mathematical analysis of the model

5.1. Basic Reproduction Number ($\mathcal{R}_0$)

To derive the $\mathcal{R}_0$ of the model, we define X_s as the set of all disease-free states,

$$X_s = \{x \geq 0 | x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^T$, such that $x_i \geq 0$ represents the number of individuals in each compartment, and each function is assumed to be continuously differentiable at least twice in each variable (C^2) [38].

Rearranging the equations so that the first m contain the infected individuals, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ be the rate of individuals entering compartment i by other means, and $\mathcal{V}_i^-(x)$ be the rate of individuals leaving compartment i . The disease transmission model consists of non-negative initial conditions together with the following system of equations:

$$\dot{x} = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), i = 1, \dots, n,$$

where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. Here, $F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right]$ and $V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right]$, where x_0 is a Disease-Free Equilibrium (DFE), and $1 \leq i, j \leq m$, and $\mathcal{R}_0 = \rho(FV^{-1})$.

The formulation for $\mathcal{R}_0$ in both model, as well as the complete SIR/SEI system is given by:

SIR: In this case, $m = 1$, and the compartments are arranged as $[I_H, S_H, R_H]$. Since $\mathcal{R}_0$ is calculated with normalized values, the necessary equations are multiplyed by N to remove the denominator. Specifically, for the SIR case, as R_H is not used in any of the equations, it can be expressed solely in terms of S and I . Therefore:

- $\mathcal{F}_i(x)$: rate of entrance of infected individuals in compartment i
- $\mathcal{V}_i(x)^-$: rate of leaving compartment i
- $\mathcal{V}_i(x)^+$: rate of entering compartment i

So we have

$$\mathcal{F} = [ab_2 I_M S_H]$$

$$\mathcal{V}^- = [\gamma I_H]$$

$$\mathcal{V}^+ = [0]$$

$$\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)^+$$

Therefore,

$$\mathcal{V} = [I_H \gamma]$$

Hence

$$F = \frac{\partial \mathcal{F}}{\partial I_M} = [ab_2 S_H]$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H} = [\gamma]$$

At the equilibrium, $[S_H^*, I_H^*] = [1, 0]$, so

$$F = [ab_2], \quad V = [\gamma] \text{ and } \mathcal{R}_0 = \left| \frac{ab_2}{\gamma} \right|.$$

SEI: In this case, $m = 2$, and the compartments are arranged as $[E_M, I_M, S_M]$. Again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$F = \begin{bmatrix} ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$\begin{aligned} F &= \frac{\partial F}{\partial E_M, I_H} \\ &= \begin{bmatrix} \frac{\partial ab_1 I_H S_M}{\partial E_M} & \frac{\partial ab_1 I_H S_M}{\partial I_H} \\ \frac{\partial 0}{\partial E_M} & \frac{\partial 0}{\partial I_H} \end{bmatrix} \\ &= \begin{bmatrix} 0 & ab_1 S_M \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} V &= \frac{\partial V}{\partial E_M, I_M} \\ &= \begin{bmatrix} \frac{\partial E_M(\mu + b_3 + l)}{\partial E_M} & \frac{\partial E_M(\mu + b_3 + l)}{\partial I_M} \\ \frac{\partial \mu I_M - b_3 E_M}{\partial E_M} & \frac{\partial \mu I_M - b_3 E_M}{\partial I_M} \end{bmatrix} \\ &= \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix} \end{aligned}$$

At the equilibrium, $[S_M^*, E_M^*, I_M^*] = [1, 0, 0]$, so

$$F = \begin{bmatrix} 0 & ab_1 \\ 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \frac{ab_1 b_3}{(b_3 + l + \mu)\mu} \right|.$$

SIR/SEI: In this case, $m = 3$, and the compartments are arranged as $[I_H, E_M, I_M, S_H, S_M]$. Once again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$F = \begin{bmatrix} ab_2 I_M S_H \\ ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} \gamma I_H \\ E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} I_H \gamma \\ E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$F = \frac{\partial F}{\partial I_H, E_M, I_M} = \begin{bmatrix} 0 & 0 & ab_2 S_H \\ ab_1 S_M & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V = \frac{\partial V}{\partial I_H, E_M, I_M} = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

At the equilibrium,

$$[S_H^*, S_M^*, I_H^*, E_M^*, I_M^*] = [1, 1, 0, 0, 0],$$

so

$$F = \begin{bmatrix} 0 & 0 & ab_2 \\ ab_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \sqrt{\frac{a^2 b_1 b_2 b_3}{(b_3 + l + \mu) \gamma \mu}} = \sqrt{\mathcal{R}_{0SIR} \times \mathcal{R}_{0SEI}}$$

5.2. Understanding relations

Having the necessary epidemiological and environmental datasets ready for use, the next step taken was to verify the correlation between the number of cases and forest areas impacted by deforestation and fires, in addition to the correlation with temperature and precipitation as well.

Using the Pandas library on a Python environment, the correlation between monthly number of cases caused by *P. vivax* in Manaus and deforested area was found to be approximately -0.11, and 0.137 between cases and burnt area. In relation to climatic data, the correlation with temperature was found to be around 0.404, and with precipitation to be -0.365. Both correlation tables were taken in relation to monthly data.

On the other hand, taking the correlation between the incidence of cases in relation to the seasons, deforested and burnt areas, that was found to be 0.722, 0.314 and 0.818, respectively. Looking for the correlation in relation to seasons, temperature and precipitation, that was found to be 0.687, 0.839 and -0.883, respectively.

6. Results

The initial modelling, using the original equations and parameters as defined in [7] did not appear to properly reproduce the expected behavior of disease transmission especially in the case of the mosquito population, as can be seen in Figure 2A.

Before including the environmental factors on the used model (7-12), its results were compared to a model based on SIVEP data, keeping use of equations (7, 10-12), but using

$$I_H(t) = \sum_{r=\max\left(0, k-\frac{1}{\gamma}+1\right)}^{0 \leq k \leq 1825} \# \text{ of cases} \quad (29)$$

$$R_H(t) = \max\left(0, N(1 + r_H t) - S_H(t) - I_H(t)\right) \quad (30)$$

$I_H(t)$ now being the moving sum of cases over a period of 120 days, as it is the average duration of infectiousness, and $R_H(t)$ being the difference between the initial population and the susceptible and infected population at each day, with the addition of the daily births. The result can be seen in the graph below.

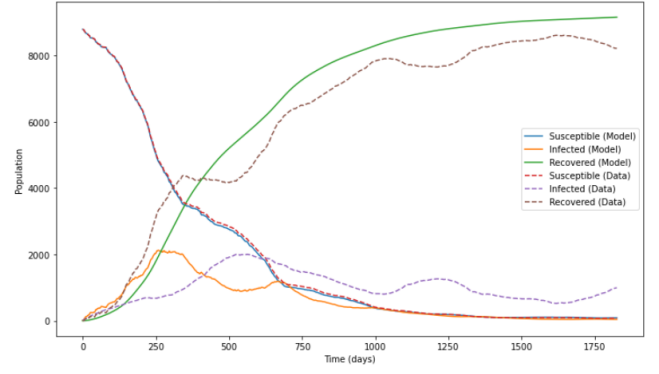


Figure 1: Comparison between observed data of infections and model estimation of I_H .

It is possible to see that the model gives a close approximation to the format of the observed number of infections and recoveries, although the infections appear to increase faster and die out given the model, a behavior that is not seen on the actual data.

At this point, a linear combination was implemented as a replacement to the use of simply T and R , in order to take into consideration the effects of deforestation and fires, with the α -function $\alpha_1 D + \alpha_2 F + \alpha_3 T + \alpha_4 R$, with $\alpha_0 = [0, 0, 1, 1]$ representing the primary version of the model.

7. Computational Implementation

All model simulations were performed using Python version 3.12.7 (conda-forge distribution). Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

The systems of time-delay ordinary differential equations (1-6) were solved numerically using the explicit Runge-Kutta method of order 8 as implemented in the `solve_ivp` function from SciPy version 1.17.1. All computations were performed on a laptop running **Windows 11** (64-bit) equipped with a **12th Gen Intel Core i5-1235U** processor (10 cores, 12 threads, 1.30 GHz base frequency) and **16 GB RAM**.

The source code and data required to reproduce all figures are available at https://github.com/RaphaLevy/Study_on_malaria_dynamics_v2.

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A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$a(T)$	Biting rate (days ⁻¹)	$(T - T') / D_1$
$\mu(T)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T))$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD / (T - T_{min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1 / (c_1 T + c_2)$
$p(T)$	Daily mosquito survival rate	$e^{(-1 / (AT^2 + BT + C))}$
$p_L(R)$	Probability of larval survival dependent on rainfall	$(4p_{ML} / R_L^2) R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1 T + c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R) p_L(T)$
$l(\tau_M)(T)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T)^{\tau_M(T)}$
$M(t)$	Total population of mosquitos	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + I_H(t) + R_H(t)$

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection	Dimensionless
γ	1/Average duration of infectiousness in humans	days ⁻¹
T_1	Mean temperature in the absence of seasonality	°C
T_2	Amplitude of seasonal variability in temperature	Dimensionless
ω_1	Angular frequency of seasonal oscillations in temperature	months ⁻¹
ϕ_1	Phase lag of temperature variability (phase shift)	Dimensionless
R_1	Average monthly precipitation in the absence of seasonality	mm
R_2	Amplitude of seasonal variability in precipitation	Dimensionless
ω_2	Angular frequency of seasonal oscillations in precipitation	months ⁻¹
ϕ_2	Phase lag of precipitation variability (phase shift)	Dimensionless
B_E	Number of eggs laid per adult per oviposition	Dimensionless
p_{ME}	Maximum probability of egg survival	Dimensionless
p_{ML}	Maximum probability of larval survival	Dimensionless
p_{MP}	Maximum probability of pupal survival	Dimensionless
τ_E	Duration of the egg development phase	days
τ_P	Duration of the pupal development phase	days
τ_H	Latent period of infection within humans	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 [8]	mm
T_{min}	Minimum temperature, below which there is no development of the parasite: 14.5	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [9, 10]	°C days
A	Empirical sensitivity parameter	(°C ² days) ⁻¹
B	Empirical sensitivity parameter	(°C days) ⁻¹
C	Empirical sensitivity parameter	days ⁻¹
D_1	Temperature scaling factor of the biting rate	°C days
c_1	Empirical sensitivity parameter	(°C days) ⁻¹
c_2	Empirical sensitivity parameter	days ⁻¹
T'	Biting rate temperature threshold ¹	°C
b_3	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
r_H	Net daily recruitment rate per capita	days ⁻¹

¹ In Parham and Michael [7], this parameter is denoted T_1 . We rename it here to avoid confusion with the mean temperature in the absence of seasonality, also denoted T_1 in the original paper.

A.2. Figures

Malaria SIR Model with Parameters: $I_{H0} = 1000$, $E_{H0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{130}$, $R_L = 50$, $T^* = 19.9$

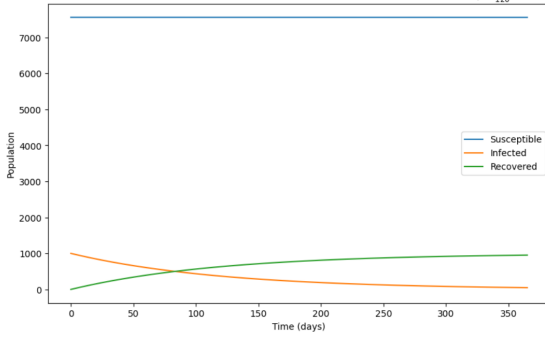


Figure 1A: Results of the initial SIR model, with equations and parameters from [7].

Malaria SEI Model with Parameters: $I_{H0} = 1000$, $E_{H0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{130}$, $R_L = 50$, $T^* = 19.9$

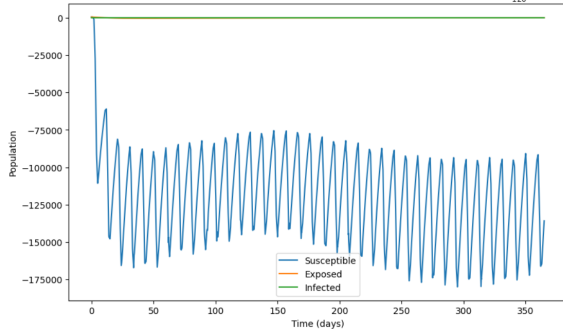


Figure 2A: Results of the initial SEI model, with equations and parameters from [7].

Malaria SEI Model with fixed probabilities and Parameters: $I_{H0} = 1000$, $E_{H0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{130}$, $R_L = 50$, $T^* = 19.9$

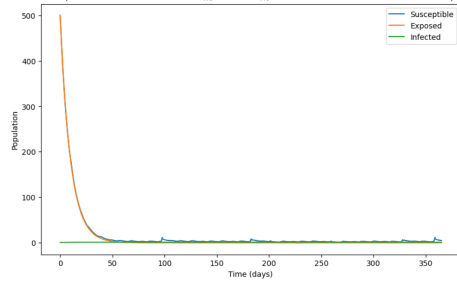


Figure 3A: Results of the initial SEI model, with equations and parameters from [7], ensuring that probabilities and rates are in $[0, 1]$.